

Doaa Alaaelden

Giza, Egypt | [Doaa Alaa](#) | 01146278595 | [Linkedin](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI Engineer, Computer Science and Artificial Intelligence graduate with strong hands-on experience in Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, and Evolutionary Algorithms. Experienced in designing intelligent systems and developing backend applications using Python, FastAPI, REST APIs, and Clean Architecture. Built AI solutions ranging from sentiment analysis and optimization systems to generative AI applications for education. Passionate about solving real-world problems through AI while continuously expanding knowledge in Generative AI, cloud computing, and scalable software engineering.

EDUCATION

Bachelor of Computer Science and Artificial Intelligence (2022 - 2026)

AI Department, Capital University

Grade: Excellent

GPA: 3.75 / 4.00

EXPERIENCE

Database Intern — ATEC August 2022 – November 2022

- Analyzed relational database schemas to identify structural inconsistencies and data integrity issues.
- Assisted in improving database design and optimizing schema organization.
- Collaborated with the database team to maintain data consistency and reliability.
- Participated in database testing and validation processes.

ICT Teacher — Azhari Academy November 2025 – May 2026

- Taught ICT concepts and computer fundamentals to students.
- Delivered practical, hands-on sessions to reinforce theoretical concepts.
- Designed and conducted quizzes and assessments to evaluate students' understanding and progress.
- Created an engaging and supportive classroom environment that encouraged active participation and learning.

CERTIFICATIONS

Data Analysis Certificate — AYB (بلدي يا عشانك)

Sponsored by CIB Foundation

Digitera Program — Tech Track Certificate of Completion — iCareer

In partnership with Plan International Egypt, under the Youth Inclusion and Employment Program (Ready for Tomorrow)

PROJECTS

EDUGINIE — Graduation Project Generative AI for Education: Personalized Learning Platform | Technologies: Python, Generative AI, NLP, LLMs

- Developing an AI-powered educational platform that converts Text, PDF, PowerPoint presentations, and Voice into personalized educational videos (VARK Based).
- Integrating Generative AI models to automate educational content generation.
- Designing an intelligent workflow for adaptive and personalized learning experiences.
- Focusing on improving student engagement through AI-generated multimedia content.

Exam Scheduling System Technologies: Python, Evolutionary Algorithms

- Developed an intelligent exam scheduling system using Evolutionary Algorithms (Genetic Algorithm) to generate optimized examination timetables.
- Implemented conflict detection and constraint handling to eliminate student and room scheduling conflicts.
- Improved scheduling efficiency through optimization and backtracking techniques.

Task Tracker Web Application Technologies: Python, FastAPI, SQLite, JWT, Clean Architecture

- Developed a task management web application following Clean Architecture principles.
- Implemented secure authentication and authorization using JWT.

- Designed RESTful APIs for task and user management.
- Built profile management, notifications, and task tracking functionalities.

Aspect-Based Sentiment Analysis*Technologies: Python, NLP, Deep Learning*

- Built an Aspect-Based Sentiment Analysis system for opinion mining.
- Applied text preprocessing, feature engineering, and deep learning techniques for sentiment classification.
- Analyzed customer opinions at the aspect level to improve sentiment interpretation.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, SQL

Artificial Intelligence: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Natural Language Understanding (NLU), Natural Language Generation (NLG), Generative AI, Neural Networks, Evolutionary Algorithms, Data Analysis

Backend Development: FastAPI, RESTful APIs, Clean Architecture, JWT Authentication

Databases: SQLite, MySQL, SQL, Oracle

Cloud: AWS Fundamentals, Amazon EC2, Auto Scaling, Elastic Load Balancing, ECS, AWS Lambda

Tools: GitHub, VS Code, Jupyter Notebook

Product & Requirements: BRD & PRD Development, Requirements Analysis, User Stories, Scope Definition, Acceptance Criteria

Agile Delivery: Jira, Sprint Planning, Backlog Management, Delivery Tracking

AI-Assisted Development: Cursor, Claude Code, MCP Workflows, AI-Assisted Coding, Debugging & Testing

Engineering Collaboration: Git, GitHub, Pull Requests, Code Reviews, Repository Workflows

Debugging & Quality: Root Cause Analysis, Bug Investigation, Verification, Regression Testing

SOFT SKILLS

Analytical Thinking, Problem Solving, Team Collaboration, Communication, Adaptability, Continuous Learning

LANGUAGES

Arabic — Native | English — Very Good